

MERN Stack Development

From Frontend to Backend

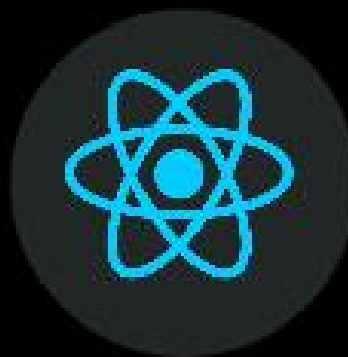
MERN STACK



MongoDB



Express JS



React JS



Node JS

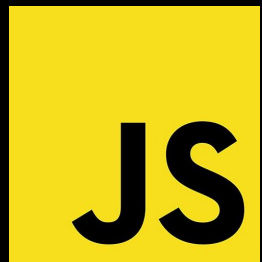
Frontend Technologies



HTML



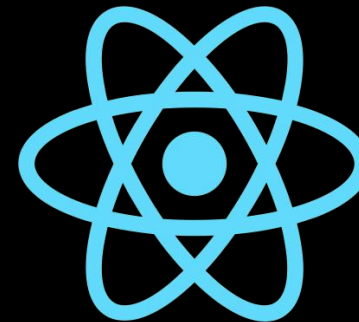
CSS



Js



Bootstrap



React

HTML (Hypertext Markup Language)

- What is HTML?
- Structure Tags
- Boilerplate code
- HTML Tags

```
<h1> to <h6> , <p> , <a> , <img> , <div> , <span> , <ul>  
<ol> , <li> , <form> , <input> , <button> , <br> , <strong>  
<em> , <table> , <tr> , <th> , <td> , <blockquote> , <q>  
<abbr> , <address> , <iframe>
```

- HTML tables
- Types of HTML Tags
- Types of Attributes in HTML
- Tag / Element / Attribute
- HTML Comments
- HTML Favicon

CSS (Cascading Style Sheets)

What is CSS?

Why Use CSS?

Three Ways to Add CSS:

Benefits of CSS

CSS Syntax

Text properties : values

CSS Selectors

1. Universal Selector
2. Element Selector
3. ID Selector
4. Class Selector
5. Grouped Selector
6. Descendant & Child Selectors
7. Attribute Selectors
8. Pseudo-classes
9. Pseudo-elements

CSS Box Model

1. Margin
2. Padding
3. Border

display property

1. inline
2. block
3. inline-block

Image float

position property

1. static
2. relative
3. absolute
4. fixed
5. sticky

Pseudo-classes in CSS

1. :active
2. :hover
3. :focus

CSS Transforms

CSS Transitions

CSS Animations

Responsive Web design

1. Media Queries
2. Flexbox
3. Grid Layout
4. Bootstarp layout system

Bootstrap

What is Bootstrap?

How to Include Bootstrap ?

How Bootstrap working ?

Why Use Bootstrap?

Text Classes

Background Color Classes

Spacing Classes (Margin & Padding)

Flexbox Utility Classes

Width & Height Classes

Bootstrap Container

Bootstrap Grid System

JavaScript

What is JavaScript?

Js importance

History

How to add .Js to the HTML

JavaScript Engine and Execution Process

Variables

Datatypes (primitive, Non-Primitive)

operators

control flow statements

Functions

Hoisting & Temporal Dead Zone (TDZ)

- Types of functions
- Function Declaration
 - Function Expression
 - Arrow Functions (ES6)
 - Anonymous Functions
 - Callback Functions
 - Recursive Functions
 - Higher-Order Functions

Scope of a variable

setTimeout(), setInterval

clousers

Excecuion Context and Call stack

Error Handling

BOM and DOM

Form data Handling

Installing **node**

Oops in Js

Events in js

APIs

js - synchronized - Asynchronized

- Callbacks
- Promises
- async and await
- Fetch Method
- Axios

React.js

- What is React
- The story
- Features of React.js
- Vite + React: Local Environment Setup
- Folder Structure (After Setup)
- JSX (JavaScript XML)
- React Components
- React Props
- import & export in React (ES6 Modules)
- Styling in React
- Rendering Arrays in React
- Conditional Rendering in React
- Events in React.js
- State, render and rerender
- state and Lifting State Up in React

Hooks in React

- useState
- useEffect
- useRef
- useContext
- useMemo
- useCallback

- Class-Based Components in React
- Forms in React
- Custom Hooks in React
- Higher-Order Component (HOC)
- Routing in Web Development
- Lazy Loading in React
- Performance Optimizing Techniques

Backend Technologies



Node.js



Express.js



MongoDB

Node.js & Express

- What is Node.js?
- The Story Behind Node.js – Ryan Dahl
- How to Install Node.js
- Running JavaScript with Node.js
- What is NPM?
- Architecture of Node.js
- process Object in Node.js
- Chrome V8 vs Node.js V8
- Modules in Node.js
- File Handling in Node.js
- Create a Simple HTTP Server using Node.js
- URL
- Difference Between HTTP and HTTPS
- HTTP Request Methods
- Handling different methods in Node Server
- Problems When Using Node.js Without Express:
- Express
- Versioning in Node.js (npm)
- REST API
- Building RESTful APIs
- Postman
- Middlewares in Express
- HTTP Headers
- HTTP Status Codes
- Connecting MongoDB to our Application
- CURD Operation with Database
- MVC Pattern

MongoDB

- What is a Database?
- What is a DBMS (Database Management System)?
- Role of Database in Client-Server Architecture
- What is ORM (Object Relational Mapping)?
- What is ODM (Object Document Mapping)?
- What is SQL and NoSQL?
- What is MongoDB and Mongoose?
- What is MongoDB?
- Installing and Setting Up MongoDB
- Database / Collection / Document
- MongoDB Structure
- Basic MongoDB commands
- Inserting Documents in MongoDB
- Querying Documents
- MongoDB Data Types
- Sorting Data in MongoDB
- MongoDB limit()
- Querying Documents
- Updating Data in MongoDB
- Deleting Data in MongoDB
- comparison operators
- Logical Operators
- Indexes
- Aggregation Framework

Projects

Frontend Projects (2 Projects)

Project 1: Responsive Food Ordering Website

- Build UI using React + Tailwind CSS
- Component-based architecture
- React Router for navigation
- State management with hooks
- Dynamic product listing and filtering

Project 2: Job Portal Frontend

- Modern UI with reusable components
- Form handling and validation
- API integration
- Authentication UI (Login / Signup)
- Responsive design for all devices

Backend Projects (2 Projects)

Project 3: REST API for Task Management

- Build APIs using Node.js and Express.js
- CRUD operations
- MongoDB database integration
- Authentication using JWT
- API testing with Postman

Project 4: Authentication System

- User Registration & Login APIs
- Password encryption using bcrypt
- Token-based authentication
- Middleware and protected routes

Complete MERN Stack Application

Frontend (React.js)

- Build responsive UI using React
- Component-based architecture
- Routing using React Router
- Form handling and validation
- API integration using Fetch / Axios
- State management with React Hooks

Backend (Node.js + Express.js)

- Develop RESTful APIs
- Implement CRUD operations
- Middleware and routing
- User authentication using JWT
- Password encryption using bcrypt
- API testing using Postman

Project Outcome

Students gain hands-on experience in building, integrating, and deploying a complete production-ready MERN application, creating a strong portfolio project for job opportunities.

Database (MongoDB)

- Design schemas using Mongoose
- Perform CRUD operations
- Data validation and relationships
- Store and manage application data

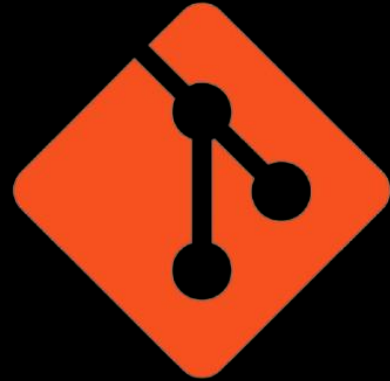
Authentication & Security

- User Registration and Login
- JWT-based authentication
- Protected routes
- Secure password storage

Deployment

- Frontend deployment (Vercel / Netlify)
- Backend deployment (Render / VPS)
- Cloud database using MongoDB Atlas

Version Control



Git



GitHub

Hosting Applications

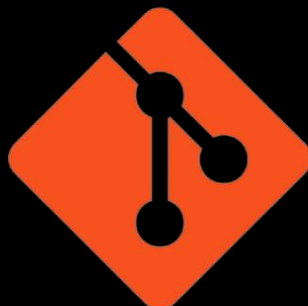
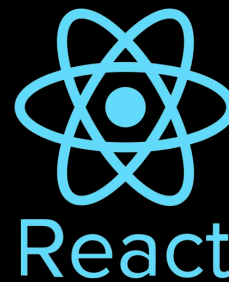
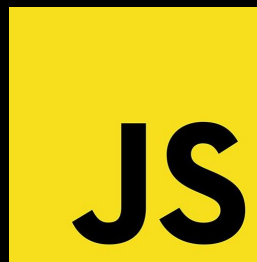


GitHub

Vercel



Tools & Technologies Covered



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